of an imminent collision at every frame of a dashcam video—is safety-critical yet difficult to scale, because collecting in-domain annotated accident footage for every deployment scenario is prohibitively expensive. We study this task under a zero-shot *setting where no target-domain training data is available: the model must learn exclusively from a publicly available binary-labelled driving-accident dataset and generalise to unseen dashcam footage. We propose a framework that bridges the gap between the frame-level temporal risk estimation task and coarsely labelled binary accident datasets by coupling a VideoMAE-v2 backbone with a per-frame prediction head under a sliding-window protocol. Our method achieves* ***2nd place*** *in the 2026 CVPR@AUTOPILOT Zero-Shot Accident Anticipation competition [1]. Code is available at* https://github.com/TimeSouth/zero-shot-taa-solution.

1. Introduction

Traffic accident anticipation (TAA) aims to estimate, at every frame of a dashcam video, the probability that a collision or near-miss will occur in the immediate future. Accurate and timely frame-level risk prediction is a prerequisite for driver-assistance alerts and autonomous emergency braking [2, 5], alongside other safety-critical perception tasks in robust autonomous driving and 3D scene understanding [6, 8, 19, 20, 23, 25]. Prior work has achieved encouraging results when in-domain annotated data is available [2, 3, 5, 22]; however, collecting and labelling dashcam accident footage for every target domain is prohibitively expensive. A more practical paradigm is *zero-shot* TAA, in which the model is trained solely on publicly available driving-accident data and must generalise to previously un-

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challenges arise. First, a *task–data granularity mismatch*: TAA requires dense, frame-level risk scores over clips of a fixed temporal format (*e.g.*, 150 frames at 30 fps), yet publicly available accident datasets provide only coarse binary labels (accident vs. normal). Bridging this granularity gap during training is essential for meaningful per-frame supervision. Second, a *long-clip / short-window modelling* problem: while advanced video representation and multimodal learning have driven significant progress across various visual tasks (e.g., action assessment, temporal localization, video retrieval, and commonsense reasoning) [10–17, 24], state-of-the-art video encoders such as VideoMAE-v2 [21] accept only 16 frames per forward pass, substantially fewer than the full clip length, necessitating a principled window-to-clip aggregation mechanism.

We address the first challenge by restructuring a publicly available binary-labelled accident dataset into temporally standardised clips with explicit positive–negative balancing, so that per-frame supervision can be derived from clip-level annotations (Sec. 2.2). We address the second by coupling a VideoMAE-v2 backbone with a per-frame prediction head under a sliding-window protocol, and recovering dense clip-level risk sequences via overlap averaging and temporal interpolation (Sec. 2.3). Furthermore, we observe that the source-trained model suffers from a systematic *posterior shift* when applied to the target domain: the relative ranking of frame-level predictions is well-preserved, yet the predicted confidence distribution is systematically compressed, preventing effective decision-making. We address this with a lightweight, training-free test-time domain adaptation module that leverages task-specific temporal priors and the model's own prediction-distribution statistics to recalibrate the output without any target-domain labels (Sec. 2.4).

Our method achieves **2nd place** in the 2026 CVPR@AUTOPILOT Zero-Shot Accident Anticipation nary label $y \in \{0,1\}$ (accident vs. normal) and the event onset time τ. During training, we derive frame-level supervision $y_t \in \{0,1\}$ from these clip-level annotations (detailed in Sec. 2.2).

Our pipeline consists of three stages: (i) *dataset curation* (Sec. 2.2), which restructures the source data into temporally standardised clips with balanced class distribution; (ii) *model training and inference* (Sec. 2.3), which fine-tunes a video encoder to predict $\{p_t\}$ via a sliding-window protocol; and (iii) *test-time domain adaptation* (Sec. 2.4), which recalibrates the output distribution at test time by exploiting task-specific temporal priors and the model's own prediction-distribution statistics, without requiring any target-domain labels.

2.2. Dataset Construction

Source. We use the public Nexar Crash / Dashcam Collision Prediction dataset [9] as the sole training and validation source. Each video is annotated with a clip-level label (collision / near-miss vs. normal driving) and the time-of-event τ.

Clip Splitting. Each competition [1] test clip is exactly 150 frames at 30 fps, so we slice every Nexar video to match. For positive videos we discard the last 2 s before τ, then slide a 5 s window over the remaining footage, so the model never observes the collision moment itself. For negative videos we slide directly and set $\tau = 250$ (well outside the clip) so that the time-weighted CE term reduces to plain CE. Frame-level labels follow $y_t = 1$ in positive clips and $y_t = 0$ otherwise.

Class Balancing. A naïve sliding-window split is heavily biased toward negative clips, on which a CE objective collapses to the trivial "always-negative" minimiser. We therefore apply a tiered balancing scheme that (i) keeps every positive clip, (ii) keeps every negative clip cut from a positive video—these provide informative *pre-/post-event* con-

tokens $Z \in \mathbb{R}^{N \times D}$. On top of the encoder we attach a lightweight per-frame classification head that fuses a global mean-pooled feature with temporally upsampled token features, producing per-frame risk probabilities $\{p_t^{(w)}\}_{t=1}^{16} \in [0,1]$ for each window. The entire model is trained end-to-end.

Training objective. For window samples with frame label $y_t \in \{0,1\}$, time-to-event τ and global frame index t, we minimise

$$\mathcal{L} = \frac{1}{2}\Big(\mathbb{E}_{y_t=1}\big[\, e^{-\max(0,\,(\tau-t-1)/\mathrm{fps})} \cdot \mathrm{CE}\big] + \mathbb{E}_{y_t=0}[\mathrm{CE}]\Big) \tag{1}$$

This is the discrete "Exp-Loss" commonly used for accident anticipation [2]: it down-weights frame predictions that are far from the event so that the model is encouraged to be confident only when the accident is temporally imminent, while still being penalised by full CE elsewhere.

Window-to-clip aggregation. For each 150-frame test clip we (i) slide a 16-frame window with stride 1 across the 10 fps-sampled sequence, (ii) run the network on every window to obtain $\{p_t^{(w)}\}$, (iii) for every sampled frame position average the predictions of all windows that cover it (*overlap averaging*), and (iv) linearly interpolate the resulting sparse probability vector back to all 150 frame indices (*temporal interpolation*). Step (iv) is necessary because the model only outputs predictions at 10 fps-sampled positions, while the evaluation requires 30 fps frame-level scores.

2.4. Test-Time Domain Adaptation

We observe that the source-trained model suffers from a *posterior shift* on the target domain: frame-level risk rankings are well-preserved, yet the absolute confidence scale is systematically compressed, a common artefact in unsupervised domain adaptation. We address this with a training-free, three-component test-time adaptation module.

Temporal confidence aggregation. In the TAA task, frames closer to the end of a clip carry stronger causal evidence about an impending event, since the accident—if any—occurs shortly after the clip ends. Motivated by this *temporal saliency prior*, we aggregate the per-frame predictions into a single clip-level confidence anchor p_{final} by retaining the last-frame prediction, which empirically exhibits the highest signal-to-noise ratio among all frame positions.
Temporal risk prior reconstruction. Accident risk possesses a well-known *monotonicity prior*: the probability of an imminent collision increases monotonically as the temporal distance to the event decreases. We exploit this inductive bias to reconstruct a dense, physically plausible frame-level risk curve from the clip-level anchor. Specifically, we parameterise the reconstruction as

$$p_t = p_{\text{start}} + (\beta\, p_{\text{final}} - p_{\text{start}}) \cdot h(t/T;\, k, \mu), \tag{2}$$

where h is a monotonically increasing temporal basis function that blends exponential and logarithmic schedules, k and μ are shape parameters estimated from the second-order statistics of the raw curve, $\beta < 1$ is a head-room factor ensuring strict monotonicity below the end anchor, and p_{start} is the initial risk level. This step effectively *propagates* the model's high-confidence late-segment signal backward along the temporal axis under a physically motivated constraint.
Prediction distribution alignment. Even after temporal reconstruction, the predicted confidence distribution on the target domain remains systematically shifted relative to the source domain's decision boundary. Since no target-domain labels are available, we perform *unsupervised recalibration* using only the moments of the model's own prediction distribution on the target set. Concretely, we compute the empirical median m of $\{p_{\text{final}}\}$ across all target clips as a domain-specific statistic, and apply an order-preserving mapping that aligns the predicted distribution to the expected decision boundary. This procedure is analogous to test-time batch normalisation in unsupervised domain adaptation: it shifts the activation statistics to match the target distribution without retraining. Crucially, because the only target-set quantity used is a moment of the model's own output distribution, the zero-shot constraint is fully respected.

3. Experiments

3.1. Implementation Details

The full model (87.1 M parameters) is fine-tuned with AdamW [7], learning rate 1×10^{-5} for the backbone and 1×10^{-4} for the head, weight decay 5×10^{-4}, batch size 3 with 8-step gradient accumulation (effective batch 24), 20 epochs, StepLR ($\gamma = 0.1$ every 5 epochs), gradient clipping with max norm 5.0, and automatic mixed precision (AMP). Training takes approximately 17 hours on a single NVIDIA V100 GPU. We select the submitted checkpoint based on the validation AP, AUC, and mTTA.
Model scale. The VideoMAE-v2 Base backbone has 86.2 M parameters; the per-frame head adds 0.9 M, totalling 87.1 M trainable parameters.
Test-time adaptation hyper-parameters. On the source validation set, the raw model produces median $p_{\text{final}} \approx 0.34$ and clip-mean ≈ 0.05. The temporal risk prior uses $k \in [7, 15]$, $\mu \in [0.2, 0.8]$, and head-room factor $\beta = 0.998$. The distribution alignment step maps clips in $[m, 0.5)$ order-preservingly into $[0.505, 0.7]$.

3.2. Evaluation Protocol

We follow the official evaluation protocol of the CVPR@AUTOPILOT Zero-Shot Accident Anticipation competition [1]. The evaluation measures detection quality via frame-level Average Precision (AP) and Area Under the ROC Curve (AUC), while early and stable risk awareness is assessed via Time-To-Accident (TTA) and Stable TTA (STTA). Higher AP and AUC indicate better detection capability, whereas larger TTA metrics reflect earlier anticipation. On our held-out Nexar [9] validation split, we report the AP and AUC metrics. Additionally, we present the competition's official weighted composite score evaluated on the unseen test set, denoted as "Private LB" in our results.

3.3. Results

Beyond the final pipeline of Sec. 2, we ran two experiments to test design choices that, *a priori*, looked promising but turned out to hurt the score. Both are reported here because their negative results directly motivate decisions in our final method. In both, the *only* change from the final pipeline is the one explicitly stated; everything else is held constant.
Late-Window Classification. Instead of feeding the model a 16-frame window over the full 5 s clip, we extract only the *last 2 s* of every clip, uniformly sub-sample 16 frames from this trailing window, and produce a single clip-level binary decision (rather than a per-frame risk sequence). The clip-level probability is then replicated across all 150 output positions. Trained with the same backbone and class-balanced split as the proposed model, this variant reaches a validation AP $= 0.66$ and AUC $= 0.69$ (Tab. 1, row B), clearly below the proposed per-frame model. On the private leaderboard the gap widens to -0.027 with respect to our final submission. The underlying reason is that, by replicating a single scalar across all 150 positions, the late-window output has no temporal resolution by construction, so the post-processor of Sec. 2.4—which is order-preserving—cannot rebuild a rising risk curve from a flat one. Empirically, this confirms that *discarding the first 3 s of context costs more than the simpler late-time supervision saves*; we therefore

	Method	AP	AUC	Private LB
A	Proposed	**0.83**	**0.86**	**2.46234**
B	Late-Window	0.66	0.69	2.43530
C	Mixed w/o balance	0.78	0.60	2.35424

Table 1. Comparisons on the Nexar validation split. "AP" and "AUC" are computed on the held-out Nexar validation set, with AP reported as the standard average precision in $[0, 1]$. "Private LB" reports the corresponding submission's score on the official Kaggle *private* leaderboard of the CVPR@AUTOPILOT Zero-Shot Accident Anticipation competition, where higher is better.

retain the full 5 s input plus a per-frame head.

Cross-Dataset Mixed Training without Class Balancing. Motivated by the scarcity of accident clips in Nexar, we augment training with a second public accident dataset [22] and use the union as the new training set. Crucially, in this experiment we *bypass the class-balancing step of* Sec. 2.2: every positive clip from both sources is kept, and only Nexar's purely-negative videos are sub-sampled. The resulting training set contains substantially more positives than negatives. The model trained under this regime degenerates into an "always-positive" predictor: on validation it still attains AP $= 0.78$ (Tab. 1, row C), because AP on an imbalanced set rewards any model that tends to predict the majority class, but the corresponding AUC $= 0.60$—close to the 0.5 chance level—reveals that the model has essentially lost the ability to rank positives above negatives. The private-leaderboard score drops to 2.35, a full -0.108 gap below our final submission. This is a textbook consequence of class imbalance at the dataset level: when the marginal $P(y = 1)$ in training is pushed close to one, the cross-entropy minimiser is the constant-positive predictor, and the time-weighting term in the Exp-Loss—which only re-scales the *positive* loss—provides no counterweight. It is for exactly this reason that we make dataset-construction-time class balancing a first-class component of our pipeline rather than a tunable hyper-parameter.

4. Conclusion

We presented a zero-shot traffic accident anticipation framework that couples a VideoMAE-v2 sliding-window classifier with class-balanced dataset construction and a training-free test-time domain adaptation module. Ablation studies confirm that both full-context per-frame modelling and dataset-level class balancing are essential to avoid performance collapse. The system achieves 2nd place in the 2026 CVPR@AUTOPILOT Zero-Shot Accident Anticipation competition [1].

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